Math 526, F09 **Name:** Answer
Quiz 11 **SSN: xxx-xx-**

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (20 points) Let $\mathbf{A} = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$.

(a) Solve the linear system $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ by **Cramer's Rule.**

Solution:
$$\det A = (-1)\cdot\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} + 3\cdot\begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix}$$
$$= 1 + 3 = 4$$
$$\det B_1 = \begin{vmatrix} 1 & 1 & 3 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix} = -1$$
$$\det B_2 = \begin{vmatrix} 0 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 0 & 1 \end{vmatrix} = 1$$
$$\det B_3 = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 2 & 1 & 0 \end{vmatrix} = 1$$
$$\Rightarrow x_1 = \frac{\det B_1}{\det A} = -\frac{1}{4},\quad x_2 = \frac{\det B_2}{\det A} = \frac{1}{4},\quad x_3 = \frac{\det B_3}{\det A} = \frac{1}{4}$$

(b) Find $\mathbf{A}^{-1}$ using the cofactor formula $\mathbf{C}^T/det\mathbf{A}$.

Solution:
$$C = \begin{bmatrix} -1 & 1 & 1 \\ 2 & -6 & 2 \\ 1 & 3 & -1 \end{bmatrix}$$
$$\Rightarrow A^{-1} = \frac{C^T}{\det A} = \begin{bmatrix} -\frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & -\frac{3}{2} & \frac{3}{4} \\ \frac{1}{4} & \frac{1}{2} & -\frac{1}{4} \end{bmatrix}$$